Tristen Chen

+1 (647) 471-3631 | tristen.chen@mail.utoronto.ca | linkedin.com/in/tristenmech2t8

EDUCATION & AWARDS

University of Toronto - *B.A.Sc. in Mechanical Engineering (PEY Co-op)* Expected June 2029

- **Awards & Certificates:** Dean's Honour List, Wallace Chalmers Engineering Design Award, SolidWorks CSWA, Basic Machining Certificate, 1st in Battle of the Schools Hackathon (Robotics Simulation track).

EXPERIENCE

Optomechanical Engineer - *UTAT Space System, University of Toronto* Jan 2026–Present

- Engineered a CubeSat venting system under strict spatial constraints. Performed pressure calculations to ensure a safety factor of 3 against maximum allowable pressure.
- Redesigned an internal leaf spring damping mechanism to withstand launch vibrations. Performed structural analysis to satisfy load constraints and prevent fatigue failure.
- Managed a multi-user CAD environment and version control through Sibe PDM, collaborating with a 20-person team to maintain assembly integrity.

Team Lead - *EWB Smart Watering System, University of Toronto* Oct 2025-April 2026

- Designed and integrated an autonomous watering system by incorporating ESP32, pumps, tubing, mechanical housing, and microcontrollers with C/C++.
- Coordinated communication between technical branches, set timelines, and managed engineering dependencies to keep teams aligned and deliver project milestones for the end-of-year conference.

Automotive Manufacturing Intern - *Taiwan Isuzu Motors Co., Ltd* May 2025-Aug 2025

- Participated in semi-knocked-down (SKD) manufacturing workflows, interpreting complex assembly drawings and conducting rigorous quality control inspections on the production line.
- Analyzed production logistics, inventory storage, and assembly scheduling, evaluating facility floor plans and timetables to understand the complete manufacturing lifecycle for imported components.

PROJECTS

Project Lead - *Stroke Bot (Battle of the Schools Hackathon)* Sept 2026

- Developed an automated simulation pipeline for a 7-DOF robotic arm, securing 1st place out of 15 teams by successfully translating OpenCV image contours into precise mechanical joint poses for vector drawing.
- Engineered Python scripts (NumPy, SciPy) to automate inverse kinematics and trajectory planning, resolving spatial redundancies to simulate complex robotic toolpaths accurately.

Project Founder - *Soft Robotics Telepresence System* April 2026-Sept 2026

- Founded and led the full-cycle prototyping of a soft-robotic telepresence system, coordinating mechanical, electronic, and software development across a 3-person multidisciplinary team. Designed and prototyped a compact 6:1 planetary gearbox in SolidWorks for NEMA 17 stepper motors. Selected bearings, optimized gear profiles, and specified tolerances for the 3D-printed assemblies.
- Identified Euler buckling in an initial 3D-printed leaf spring through physical testing. Validated a compliant joint redesign utilizing SolidWorks FEA and hand-calculated stress and fatigue analysis.
- Optimized the spatial footprint by designing a double-layer motor stack and Bowden cable routing, allowing tendons to pass through the shoulder joint's center of rotation to reduce path-length variation and reduce horizontal footprint by 25%.
- Implemented a real-time hand-tracking pipeline using Python and MediaPipe to capture user gestures and translate them into robotic motion, achieving a latency of under 500ms.

Project Co-Lead - *MIE243 Cinematography Robotic Arm Design Project* Sept 2025-Dec 2025

- Defined kinematic requirements, physical load cases, and safety features for a 5-DOF arm through benchmarking against cinematography and industrial robotic systems and applying relevant ISO standards.
- Led the full-cycle mechanical design of a motorized mounting interface. Delivered a 20+ part CAD assembly, GD&T manufacturing drawings, and a detailed BOM

SKILLS

Tools: SolidWorks (CSWA), Fusion 360, Inventor, Finite Element Analysis.
Hardware & Manufacturing: Machining (Turning/ Milling), FDM 3D printing, ESP32, Circuit Prototyping, Soldering.
Mechanical Design: GD&T, BOM Generation, Gearbox Design, Component Sourcing.
Programming: Python, MediaPipe, MATLAB, C/C++, OpenCV, NumPy, SciPy.